Quantum Mechanics Questions

Wavefunctions, Probability, Stationary States, and the Infinite Square Well

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A particle of mass m has the wave function

$$\Phi(x,t) = B\, e^{-\beta\left(\frac{mx^2}{\hbar}+2it\right)},$$

where $\beta > 0$ is a real constant.

(a) Find the normalization constant B.

Answer: *Normalize the state with*

$$\int_{-\infty}^{\infty} |\Phi(x,t)|^2\, dx = 1.$$

Since

$$|\Phi|^2 = |B|^2 e^{-2\beta m x^2/\hbar},$$

we get

$$1 = |B|^2 \int_{-\infty}^{\infty} e^{-2\beta m x^2/\hbar}\, dx = |B|^2 \sqrt{\frac{\pi\hbar}{2\beta m}},$$

so

$$B = \left(\frac{2\beta m}{\pi\hbar}\right)^{1/4}$$

if we choose the positive real normalization constant. The factor $e^{-2i\beta t}$ is a pure phase and does not affect normalization.

(b) What integral should be used to compute $\langle x\rangle$? Evaluate it.

Answer: *Use*

$$\langle x\rangle = \int_{-\infty}^{\infty} \Phi^*(x,t)\, x\, \Phi(x,t)\, dx = \int_{-\infty}^{\infty} x\, |\Phi(x,t)|^2\, dx.$$

Here $|\Phi(x,t)|^2 = |B|^2 e^{-2\beta m x^2/\hbar}$ is even in x, so the integrand $x|\Phi|^2$ is odd. Therefore the integral over $(-\infty,\infty)$ vanishes, and

$$\langle x\rangle = 0.$$

Q2. At time $t = 0$, a particle is described by

$$\Psi(x,0) = \begin{cases} D\,\dfrac{x}{c}, & 0 \le x \le c, \\ D\,\dfrac{d-x}{d-c}, & c \le x \le d, \\ 0, & \text{otherwise}, \end{cases}$$

where $0 < c < d$ and D is a normalization constant. What is the probability of finding the particle in the region $x < c$?

Answer: *The desired probability is*

$$P(x<c) = \int_0^c \left| D\frac{x}{c} \right|^2 dx = \frac{D^2 c}{3}.$$

Normalization gives

$$1 = \int_0^c \left| D\frac{x}{c} \right|^2 dx + \int_c^d \left| D\frac{d-x}{d-c} \right|^2 dx = \frac{D^2 c}{3} + \frac{D^2(d-c)}{3} = \frac{D^2 d}{3},$$

so $D^2 = 3/d$. Therefore

$$P(x<c) = \frac{D^2 c}{3} = \frac{c}{d}.$$

Q3. A quantum state is written as

$$\Phi_n(x,t) = \varphi_n(x) e^{-i\mathcal{E}_n t/\hbar}.$$

Explain why this represents a stationary state by discussing its probability density.

Answer: *This is a stationary state because the time dependence appears only as the phase factor $e^{-i\mathcal{E}_n t/\hbar}$. Its probability density is*

$$|\Phi_n(x,t)|^2 = \Phi_n^*(x,t)\Phi_n(x,t) = |\varphi_n(x)|^2 e^{i\mathcal{E}_n t/\hbar} e^{-i\mathcal{E}_n t/\hbar} = |\varphi_n(x)|^2,$$

which does not depend on time. Since all measurable position probabilities remain unchanged, the state is stationary.

Q4. Consider the general expansion of a normalized quantum state in an energy eigenbasis,

$$\boldsymbol{\Psi(x,t) = \sum_{k=1}^{\infty} a_k\, u_k(x) e^{-i\varepsilon_k t/\hbar}.}$$

Separately, consider a particle of mass M in a one-dimensional rigid box of width L with potential

$$\boldsymbol{V(x) = \begin{cases} 0, & 0 \le x \le L, \\ \infty, & \text{otherwise.} \end{cases}}$$

For the box problem, use $q^2 = 2ME/\hbar^2$.

(a) What is the physical meaning of $|a_k|^2$?

Answer: *In the expansion*

$$\Psi(x,t) = \sum_k a_k u_k(x) e^{-i\varepsilon_k t/\hbar},$$

the quantity $|a_k|^2$ is the probability of obtaining the energy eigenvalue ε_k when the energy is measured. It is also the fractional weight of the kth energy eigenstate in the normalized state.

(b) Write the expectation value of the energy, $\langle H \rangle$, in terms of $|a_k|^2$.

Answer: *Because the states $u_k(x)$ are energy eigenstates of $\hat{H}$, the expectation value is the probability-weighted average of the allowed energies:*

$$\langle H \rangle = \sum_{k=1}^{\infty} |a_k|^2 \varepsilon_k.$$

This follows directly from expanding the state in the energy eigenbasis.

(a) Using $q^2 = 2ME/\hbar^2$, write the time-independent Schrödinger equation inside the box and give its general solution.

Answer: *Inside the box, $V(x) = 0$, so the time-independent Schrödinger equation becomes*

$$-\frac{\hbar^2}{2M}\frac{d^2\psi}{dx^2} = E\psi,$$

or equivalently

$$\frac{d^2\psi}{dx^2} + q^2\psi = 0.$$

Its general solution for $0 \le x \le L$ is

$$\psi(x) = A\sin(qx) + B\cos(qx),$$

while outside the box the wavefunction is zero because the potential is infinite.

(b) Use the boundary conditions to determine the allowed energies.

Answer: *The infinite walls require*

$$\psi(0) = 0 \quad \textit{and} \quad \psi(L) = 0.$$

From $\psi(0) = 0$ we obtain $B = 0$, and then $\psi(L) = 0$ gives

$$\sin(qL) = 0 \Rightarrow qL = n\pi, \qquad n = 1, 2, 3, \ldots$$

Hence

$$q_n = \frac{n\pi}{L}, \qquad E_n = \frac{\hbar^2 q_n^2}{2M} = \frac{n^2\pi^2\hbar^2}{2ML^2}.$$

(c) Find the normalized stationary-state wavefunctions.

Answer: *Using the allowed values of q_n, the eigenfunctions are proportional to $\sin(n\pi x/L)$. Imposing*

$$\int_0^L |\psi_n(x)|^2\, dx = 1$$

gives the normalization constant $\sqrt{2/L}$, so